Bocconi

CORPORATE FINANCE 30017

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Net Present Value and Internal Rate of Return

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Disclaimer

— You have all tools needed for *Ocean Carriers* after today's class.

Agenda

NET PRESENT VALUE

INTERNAL RATE OF RETURN

PRACTICAL TIPS (FOR THE CASE)

NET PRESENT VALUE

Valuing an Investment Opportunity

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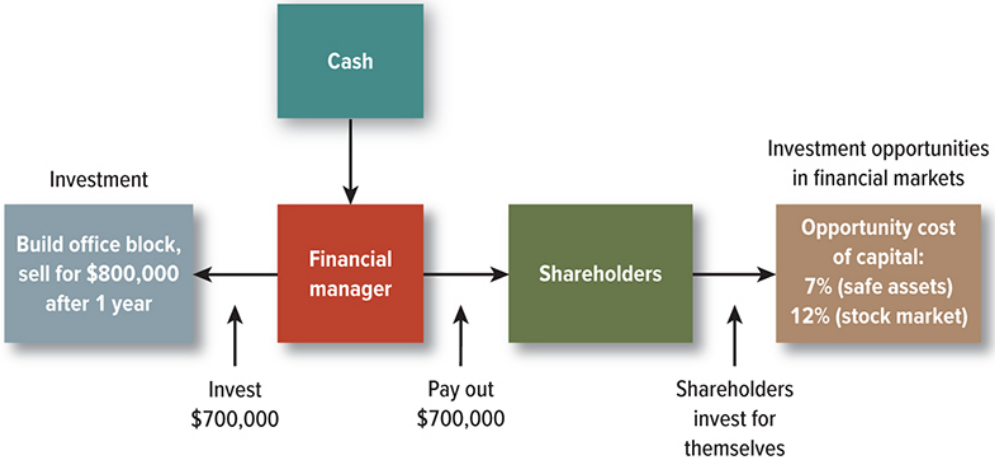


- Q1: What is the CFO's decision problem here?
- Q2: How *should* the decision be made?
- Q3: Why might the decision be made differently in practice?

Net Present value: Two period case

— *Time periods: 0, 1*

$$NPV = -I_0 + \frac{CF_1}{(1 + \boldsymbol{r})}$$

— *The discount rate solves two problems at the same time:*

1. Time-value of money: A dollar today ≠ A dollar tomorrow
2. Risk: A risky dollar ≠ A safe dollar

Net Present value: Multi period case

General formula:

$$NPV = -I_0 + \sum_{t=1}^{T} \frac{CF_t}{(1+\boldsymbol{r})^t}$$

Application: UK Consols

- How to price a UK Consol (=perpetual bond) with face value GBP 1 mio. that pays 3% coupon?
  - ...if 30 year treasuries yield 5% annually?

- Use NPV formula and the following property of infinite geometric sums:

$$\sum_{n=0}^{\infty} az^n = \frac{a}{1-z}$$

- *If you struggle with this, revisit the* Pre Course Lecture Notes *on perpetuities and annuities.*

Continuous discounting (and compounding)

- Appropriate discount rates depend on *frequency* of cashflows
  - Compounding analogue: Semi-annual interest rates are lower than annual for same interest payment
  - In the limit, aka *continuous discounting*, the formula becomes

$$NPV = \int_0^T CF(t)e^{-r_c t}dt - I_0$$

- Translating between annual and continuous discount rates:

$$r_c = \ln(1 + r)$$

- *If you feel lost, revisit* Pre Course Lecture Notes.

Something you can use Excel for:

— From our textbook:

Discounting Cash Flows

Spreadsheet programs such as Excel provide built-in functions to solve discounted cash flow (DCF) problems. You can find these functions by pressing *fx* on the Excel toolbar. If you then click on the function that you wish to use, Excel asks you for the inputs that it needs. At the bottom left of the function box there is a Help facility with an example of how the function is used.

Here is a list of useful functions for DCF problems and some points to remember when entering data:

- **FV:** Future value of single investment or annuity.
- **PV:** Present value of single future cash flow or annuity.
- **RATE:** Interest rate (or rate of return) needed to produce given future value or annuity.
- **NPER:** Number of periods (e.g., years) that it takes an investment to reach a given future value.
- **PMT:** Amount of annuity payment with a given present or future value.
- **NPV:** Calculates the value of a stream of negative and positive cash flows. (When using this function, note the warning below.)
- **EFFECT:** The effective annual interest rate, given the quoted rate (APR) and number of interest payments in a year.
- **NOMINAL:** The quoted interest rate (APR) given the effective annual interest rate.

Three warnings:

1. PV is the amount that needs to be invested today to produce a given future value. It should therefore be entered as a negative number. Entering both PV and FV with the same sign when solving for RATE results in an error message.
2. Always enter the interest or discount rate as a decimal value (for example, .05 rather than 5%).
3. Use the NPV function with care. Better still, don't use it at all. It gives the value of the cash flows one period *before* the first cash flow and not the value at the date of the first cash flow.

Something you can use Excel for:

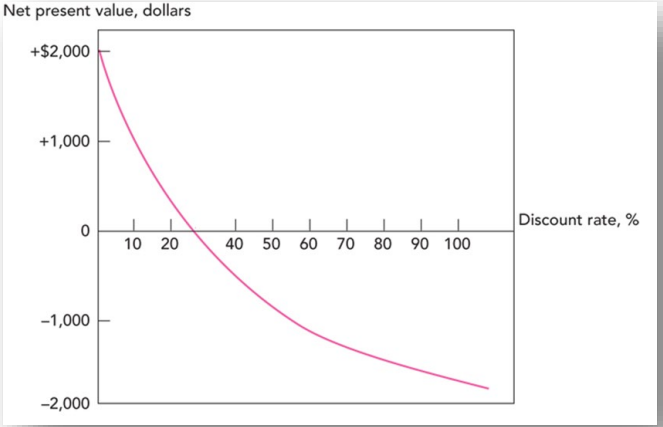
In 1880, five aboriginal trackers were each promised the equivalent of 100 Australian dollars for helping to capture the notorious outlaw Ned Kelly. One hundred and thirteen years later, the granddaughters of two of the trackers claimed that this reward had not been paid.

If the interest rate over this period averaged about 4.5%, how much would the A$100 have accumulated to? *Can use Excel's "FV".*

Something you can use Excel for:

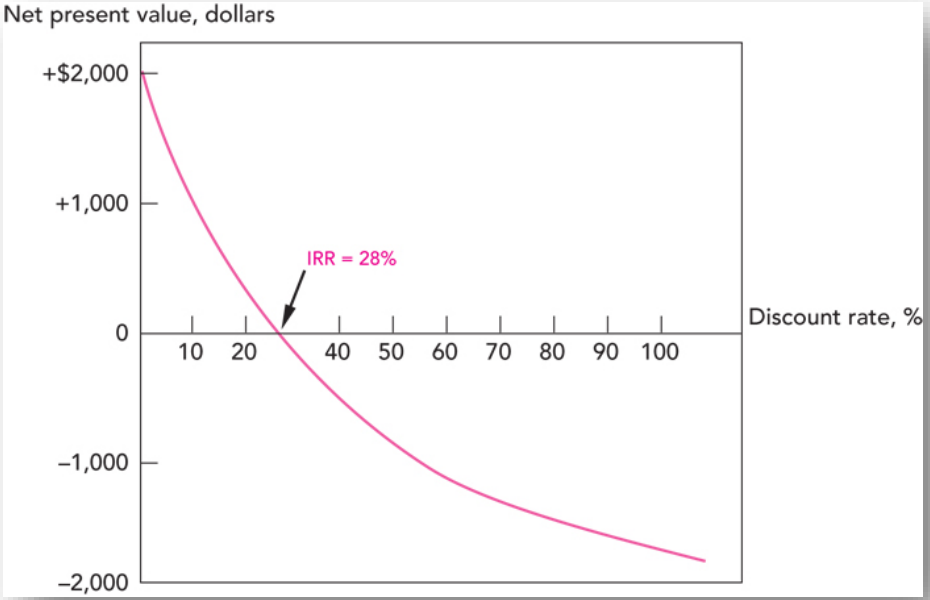
You can purchase a turbo powered machine tool gadget for $4,000. The investment will generate $2,000 and $4,000 in cash flows for two years, respectively.

Generate an NPV plot like the one below, by varying the discount rate. *Use Excel's NPV.*



INTERNAL RATE OF RETURN

Definition



— *Can verify with Excel's IRR.*

IRR Pathologies: Borrowing fallacy

- With some cash flows (as noted below) the NPV of the project **increases** as the discount rate increases.
- This is contrary to the normal relationship between NPV and discount rates.

Project	C_0	C_1	IRR	$NPV\ @10\%$
A	$-1{,}000$	$+1{,}500$	$+50\%$	$+364$
B	$+1{,}000$	$-1{,}500$	$+50\%$	-364

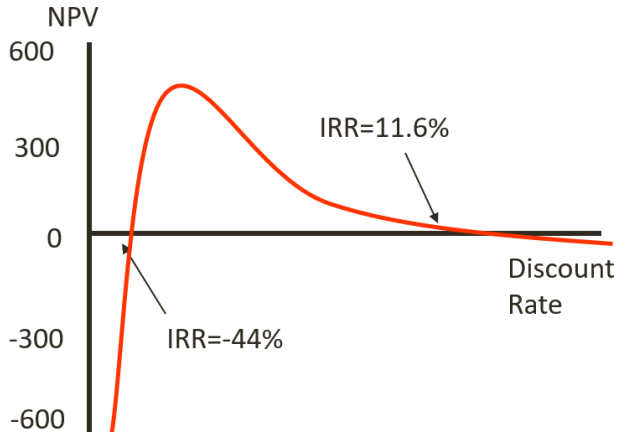
$$A: -1000 + \frac{1500}{1+r} = 0 \Leftrightarrow r = \frac{1500}{1000} - 1 = 0.5$$

$$B: +1000 - \frac{1500}{1+r} = 0 \Leftrightarrow r = \frac{1500}{1000} - 1 = 0.5$$

IRR Pathologies: Multiplicity

- Certain cash flows can generate NPV=0 at two different discount rates.
- The following cash flow generates NPV=0 at both IRR of -44% and +11.6%.
  - The reason is the double change in sign of the cashflows
  - There can be as many IRRs as there are sign changes
  - Large end-of investment negative cashflows are not rare – examples?

C_0	$C_1 \ldots\ldots$	$\ldots\ldots C_9$	C_{10}
-600	120	120	-150



IRR Pathologies: Nonexistence

- It is also possible to have **NO** IRR and a positive NPV

Project	C_0	C_1	C_2	IRR	NPV @10%
C	$+1,000$	$-3,000$	$+2,500$	*None*	$+339$

- Q: How does the NPV function look like here?

Closing remarks on IIR

- Pathologies arise from “nonstandard” cash flow profiles
  - Be wary whenever **future cash flows are negative**!

- We assume that discount rates are stable during the term of the project.
  - …implying all funds are reinvested at the IRR.
  - Often violated in practice. (Examples?)

- Generally speaking: **Better assess project profitability via NPV**

PRACTICAL TIPS (FOR THE CASE)

NPV is all about cash flow

- Economically, we care about actual **cash flows, not (accounting) income**
  - Both are obviously related, but diverge often enough

Depreciation is not a cash flow, but…

- Accounting regulation (e.g., IFRS, GAAP) treats depreciation (and amortization) as expense
  - As such they are cash-flow neutral in principle
  - However, they effectively reduce accounting income (i.e., EBIT)
  - Accounting income is basis for computing corporate income tax liabilities
  - Hence, depreciation (although expense) can have *cash flow* implications and change NPVs
- Accounting rules prescribe possible depreciation patterns
  - Limited flexibility (not: no flexibility!)

Working capital is not a cash flow, but…

- Net working capital = current assets – current liabilities
  - Balance sheet (stock) item
- But *changes* in net working capital are cash flows
  - Reductions in current liabilities are cash outflows
  - Reduction in current assets are cash inflows
    - Note: increases in accounts receivable are considered *cash outflows*
      - It's not that money left; it's that money *failed to arrive* (economically equivalent)

Nominal vs. real cash flows

- The **Fisher Equation** (Irving Fisher, 1907, "*The Rate of Interest*"):

$$1 + r_n = (1 + \pi)(1 + r_r)$$

- Caution: $r_n = \pi + r_r$ is only an *approximation!* (Why actually?)

$$NPV = -I_0 + \sum_{t=1}^{\infty} \frac{real\ CF_t}{(1 + r_r)^t} = \sum_{t=1}^{\infty} \frac{real\ CF_t (1 + \pi)^t}{(1 + r_r)^t (1 + \pi)^t} = \sum_{t=1}^{\infty} \frac{nominal\ CF_t}{(1 + r_n)^t}$$

- *Bonus question: How do we handle time-varying inflation?*

Preparation for *Ocean Carriers:* BMAE 6.3

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	Year:	0	1	2	3	4	5	6	7
	Panel A Capital Investment								
1	Cash flow from investment in fixed assets	–12,000							
2	Sale of fixed assets								1,949
3	Less tax on sale								409[a]
4	Cash flow from capital investment (1 + 2 – 3)	–12,000							1,540
	Panel B Operating Cash Flow								
5	Revenues		523	12,887	32,610	48,901	35,834	19,717	
6	Cost of goods sold		837	7,729	19,552	29,345	21,492	11,830	
7	Other costs[b]	4,000	2,200	1,210	1,331	1,464	1,611	1,772	
8	Depreciation[c]		2,000	2,000	2,000	2,000	2,000	2,000	
9	Pretax profit (5 – (6 + 7) – 8)	–4,000	–4,514	1,948	9,727	16,092	10,731	4,115	
10	Tax (.21 × 9)	–840[d]	–948	409	2,043	3,379	2,254	864	
11	Profit after tax (9 – 10)	–3,160	–3,566	1,539	7,684	12,713	8,477	3,251	
12	Operating cash flow (11 + 8)	–3,160	–1,566	3,539	9,684	14,713	10,477	5,251	
	Panel C Investment in Working Capital								
13	Working capital		550	1,289	3,261	4,890	3,583	2,002	0
14	Change in working capital		550	739	1,972	1,629	–1,307	–1,581	–2,002
15	Cash flow from investment in working capital (–14)		–550	–739	–1,972	–1,629	1,307	1,581	2,002
	Panel D Project Valuation								
16	Total project cash flow (4 + 12 + 15)	–15,160	–2,116	2,800	7,712	13,084	11,784	6,832	3,542
17	Discount factor at 20%	1.0	0.833	0.694	0.579	0.482	0.402	0.335	0.279
18	Discounted cash flows (16 × 17)	–15,160	–1,763	1,944	4,463	6,310	4,736	2,288	988
19	NPV	+3,806							

Note that depreciation is straight-line, reaching 0 in year 7. Yet, terminal assets sale value is 1,949.